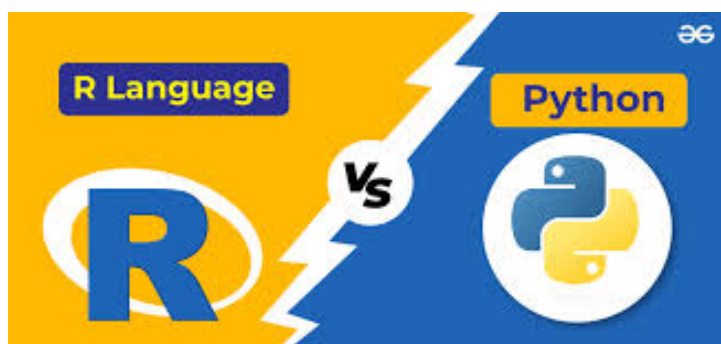


Flight Data Analysis Programing with R

Assignment Report



Student Name	Md Sariful Islam
Student ID	12310066
Department	Statistics
Course Title	Programming with R and Python
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Submitted To	Dr. Md. Siddikur Rahman (Associate Professor)
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Begum Rokeya University

Rangpur, Bangladesh

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1 Introduction

This assignment analyzes flight ticket pricing, delays, distance, flight duration, and passenger service features using the R programming language. Different statistical techniques, visualizations, hypothesis testing, correlation analysis, and regression models are applied to understand the relationships among variables.

2 Software and Libraries Used

The following R libraries were used:

- ggplot2
- dplyr
- stats

3 Data Loading and Preparation

```

1 # Set working directory
2 getwd()
3 list.files()
4 setwd("C:/Users/USER/Downloads/R & Python")
5
6 # Package path
7 .libPaths()
8 .libPaths("C:/Users/USER/Downloads/R & Python/package")
9
10 # Install and load packages
11 install.packages("ggplot2")
12 install.packages("dplyr")
13
14 library(ggplot2)
15 library(dplyr)
16
17 # Load dataset
18 flight <- read.csv("C:/Users/USER/Downloads/R & Python/flight_
    data.csv")
19
20 # Random sample

```

```
21 set.seed(66)
22 flight_sample <- flight[sample(nrow(flight), 10000), ]
```

Listing 1: Load Dataset

4 Question 1: Coach Ticket Price Analysis

4.1 Objective

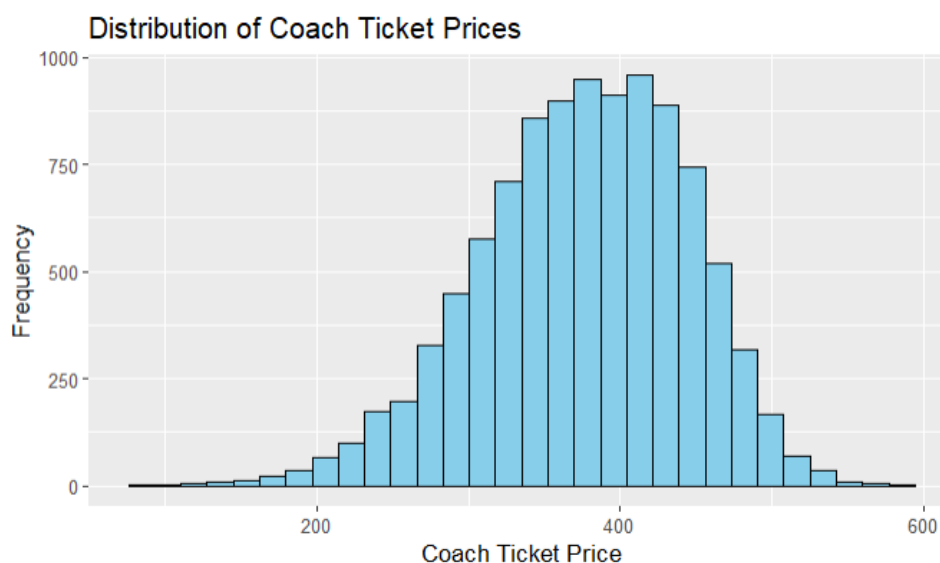
To analyze the distribution of coach ticket prices and determine whether \$500 is a reasonable ticket price.

4.2 R Code

```
1 # Summary statistics
2 summary(flight_sample$coach_price)
3
4 # Mean
5 mean(flight_sample$coach_price)
6
7 # Histogram
8 ggplot(flight_sample, aes(x=coach_price)) +
9   geom_histogram(bins=30,
10                  fill="skyblue",
11                  color="black") +
12   labs(
13     title="Distribution of Coach Ticket Prices",
14     x="Coach Ticket Price",
15     y="Frequency"
16   )
```

Listing 2: Coach Ticket Price Analysis

4.3 Output



4.4 Interpretation

The histogram and summary statistics show the overall distribution of coach ticket prices. The average price can be compared with \$500 to determine whether it is considered expensive or reasonable.

5 Question 2: Coach Prices for 8-Hour Flights

```

1 # 8-hour flights
2 flight_8 <- subset(flight_sample, hours == 8)
3
4 # Summary statistics
5 summary(flight_8$coach_price)

```

Listing 3: 8 Hour Flight Analysis

5.1 Output

5.2 Interpretation

Long-duration flights generally have higher ticket prices. Therefore, a \$500 coach ticket may be more acceptable for 8-hour flights.

6 Question 3: Distribution of Flight Delays

```

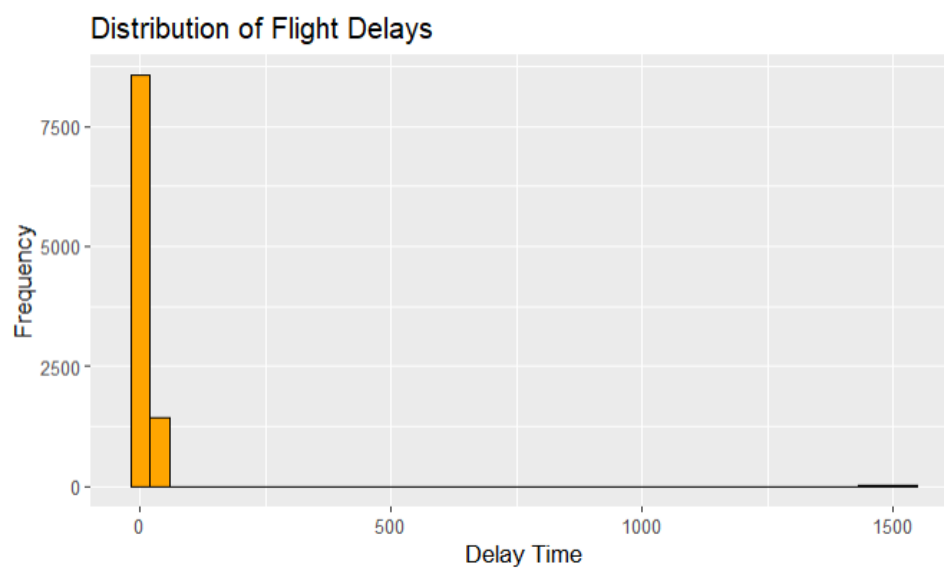
1 # Histogram
2 ggplot(flight_sample, aes(x=delay)) +
3   geom_histogram(bins=40,
4                 fill="orange",

```

```
5         color="black") +  
6     labs(  
7         title="Distribution of Flight Delays",  
8         x="Delay Time",  
9         y="Frequency"  
10    )  
11  
12 # Summary  
13 summary(flight_sample$delay)
```

Listing 4: Flight Delay Distribution

6.1 Output



6.2 Interpretation

The delay distribution may contain outliers due to extreme delays experienced by some flights.

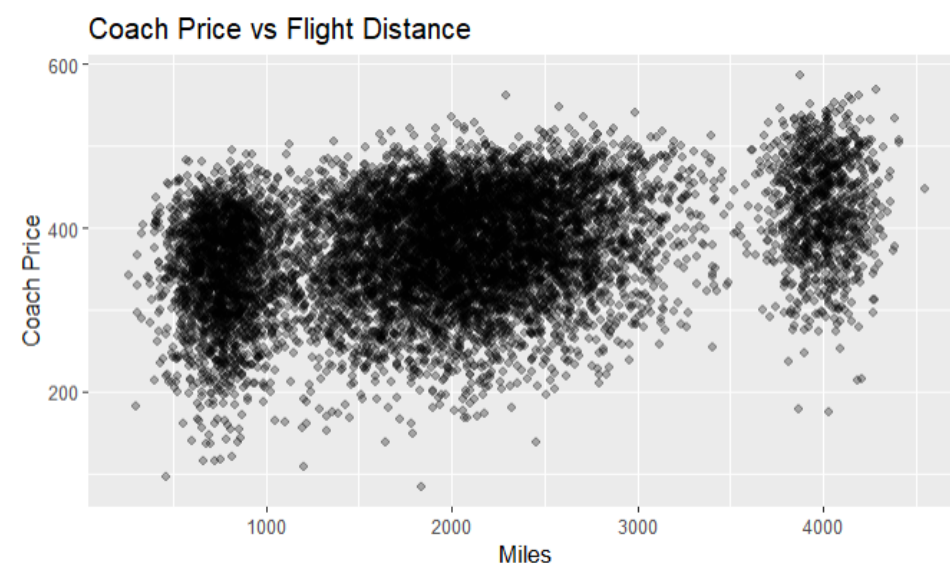
7 Question 4: Relationship Between Distance and Coach Price

```
1 # Scatter plot  
2 ggplot(flight_sample,  
3       aes(x=miles,  
4           y=coach_price)) +  
5     geom_point(alpha=0.3) +  
6     labs(  
7       title="Coach Price vs Flight Distance",  
8       x="Miles",  
9       y="Coach Price"  
10    )
```

```
11  
12 # Correlation  
13 cor(flight_sample$miles,  
14     flight_sample$coach_price)
```

Listing 5: Distance vs Coach Price

7.1 Output



7.2 Interpretation

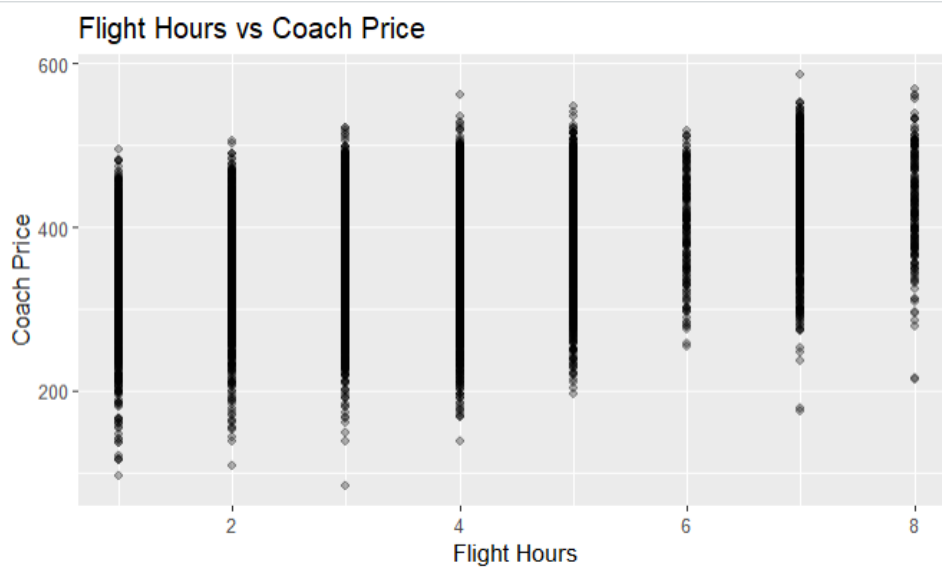
There is usually a positive relationship between flight distance and ticket price.

8 Question 5: Relationship Between Flight Hours and Coach Price

```
1 # Scatter plot  
2 ggplot(flight_sample,  
3       aes(x=hours,  
4           y=coach_price)) +  
5   geom_point(alpha=0.3) +  
6   labs(  
7     title="Flight Hours vs Coach Price",  
8     x="Flight Hours",  
9     y="Coach Price"  
10  )  
11  
12 # Correlation  
13 cor(flight_sample$hours,  
14     flight_sample$coach_price)
```

Listing 6: Flight Hours vs Coach Price

8.1 Output



8.2 Interpretation

Longer flight duration tends to increase coach ticket prices.

9 Question 6: Coach Price vs First-Class Price

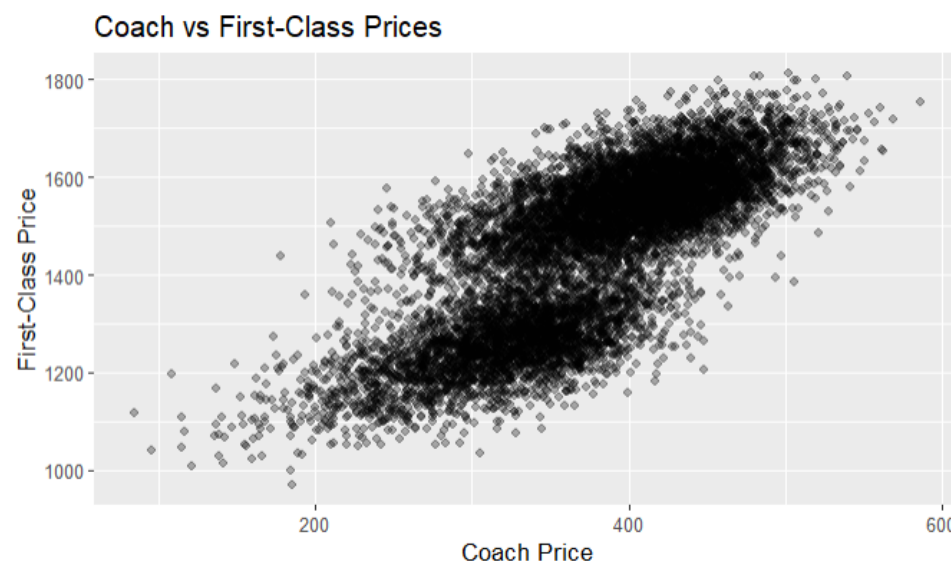
```

1 # Scatter plot
2 ggplot(flight_sample,
3       aes(x=coach_price,
4           y=firstclass_price)) +
5   geom_point(alpha=0.3) +
6   labs(
7     title="Coach vs First-Class Prices",
8     x="Coach Price",
9     y="First-Class Price"
10  )
11
12 # Correlation
13 cor(flight_sample$coach_price,
14     flight_sample$firstclass_price)

```

Listing 7: Coach vs First-Class Prices

9.1 Output



9.2 Interpretation

Higher coach prices are generally associated with higher first-class prices.

10 Question 7: In-Flight Features and Coach Ticket Prices

```
1 # Meal
2 aggregate(coach_price ~ inflight_meal,
3           data=flight_sample,
4           mean)
5
6 # Entertainment
7 aggregate(coach_price ~ inflight_entertainment,
8           data=flight_sample,
9           mean)
10
11 # Wifi
12 aggregate(coach_price ~ inflight_wifi,
13           data=flight_sample,
14           mean)
```

Listing 8: In-Flight Features

10.1 Interpretation

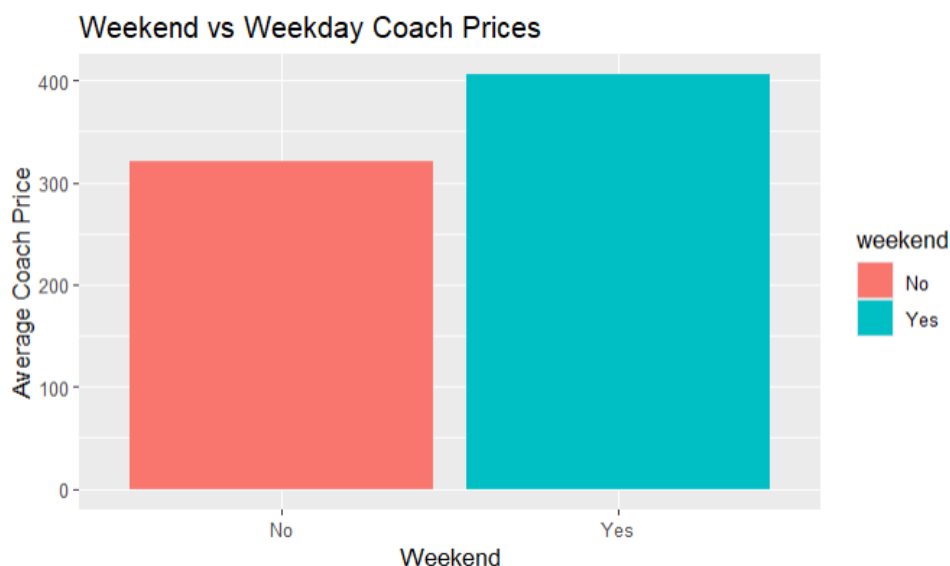
Flights with additional services such as meals, entertainment, and WiFi often have higher ticket prices.

11 Question 8: Weekend vs Weekday Ticket Prices

```
1 # Average prices
2 weekend_price <- flight_sample %>%
3   group_by(weekend) %>%
4   summarise(
5     avg_coach = mean(coach_price),
6     avg_firstclass = mean(firstclass_price)
7   )
8
9 print(weekend_price)
10
11 # Bar plot
12 ggplot(weekend_price,
13       aes(x=weekend,
14           y=avg_coach,
15           fill=weekend)) +
16   geom_bar(stat="identity") +
17   labs(
18     title="Weekend vs Weekday Coach Prices",
19     x="Weekend",
20     y="Average Coach Price"
21   )
```

Listing 9: Weekend vs Weekday Analysis

11.1 Output



11.2 Interpretation

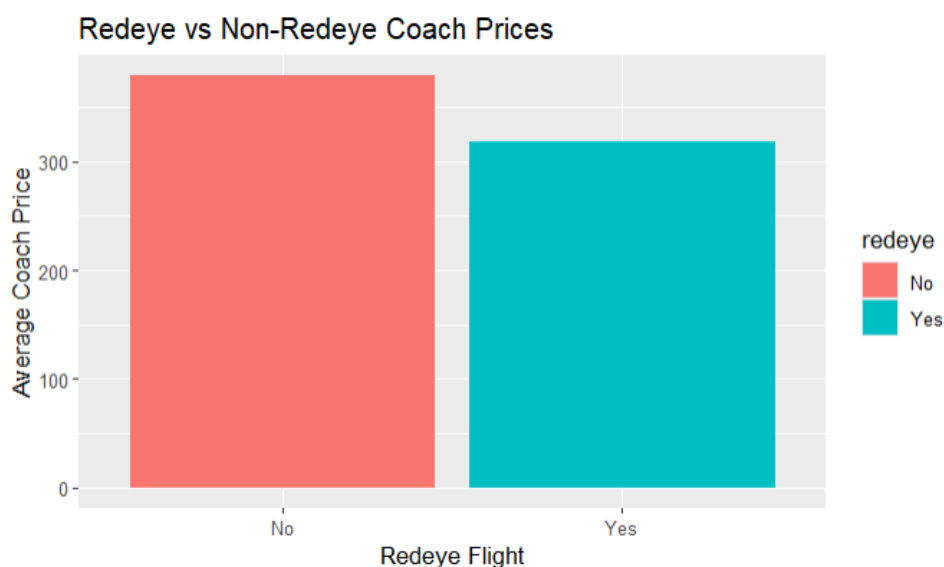
Weekend ticket prices may differ from weekday prices depending on passenger demand.

12 Question 9: Redeye vs Non-Redebye Flights

```
1 # Average prices
2 redeye_price <- flight_sample %>%
3   group_by(redeye) %>%
4   summarise(
5     avg_coach = mean(coach_price)
6   )
7
8 print(redeye_price)
9
10 # Bar plot
11 ggplot(redeye_price,
12        aes(x=redeye,
13            y=avg_coach,
14            fill=redeye)) +
15   geom_bar(stat="identity") +
16   labs(
17     title="Redeye vs Non-Redebye Coach Prices",
18     x="Redeye Flight",
19     y="Average Coach Price"
20   )
```

Listing 10: Redeye Flight Analysis

12.1 Output



12.2 Interpretation

Redeye flights are often cheaper because they operate during late-night hours.

13 Question 10: Advanced Statistical Analysis

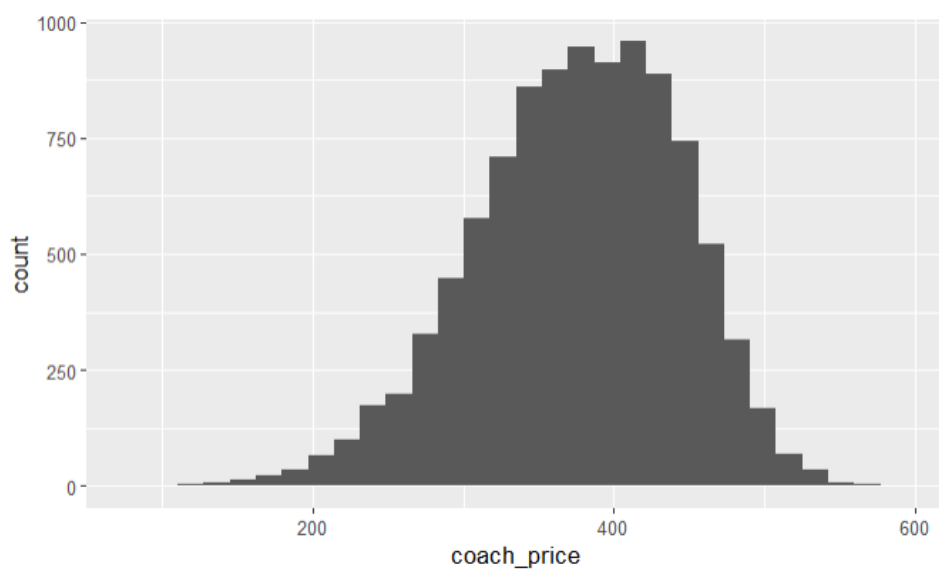
13.1 Summary Statistics

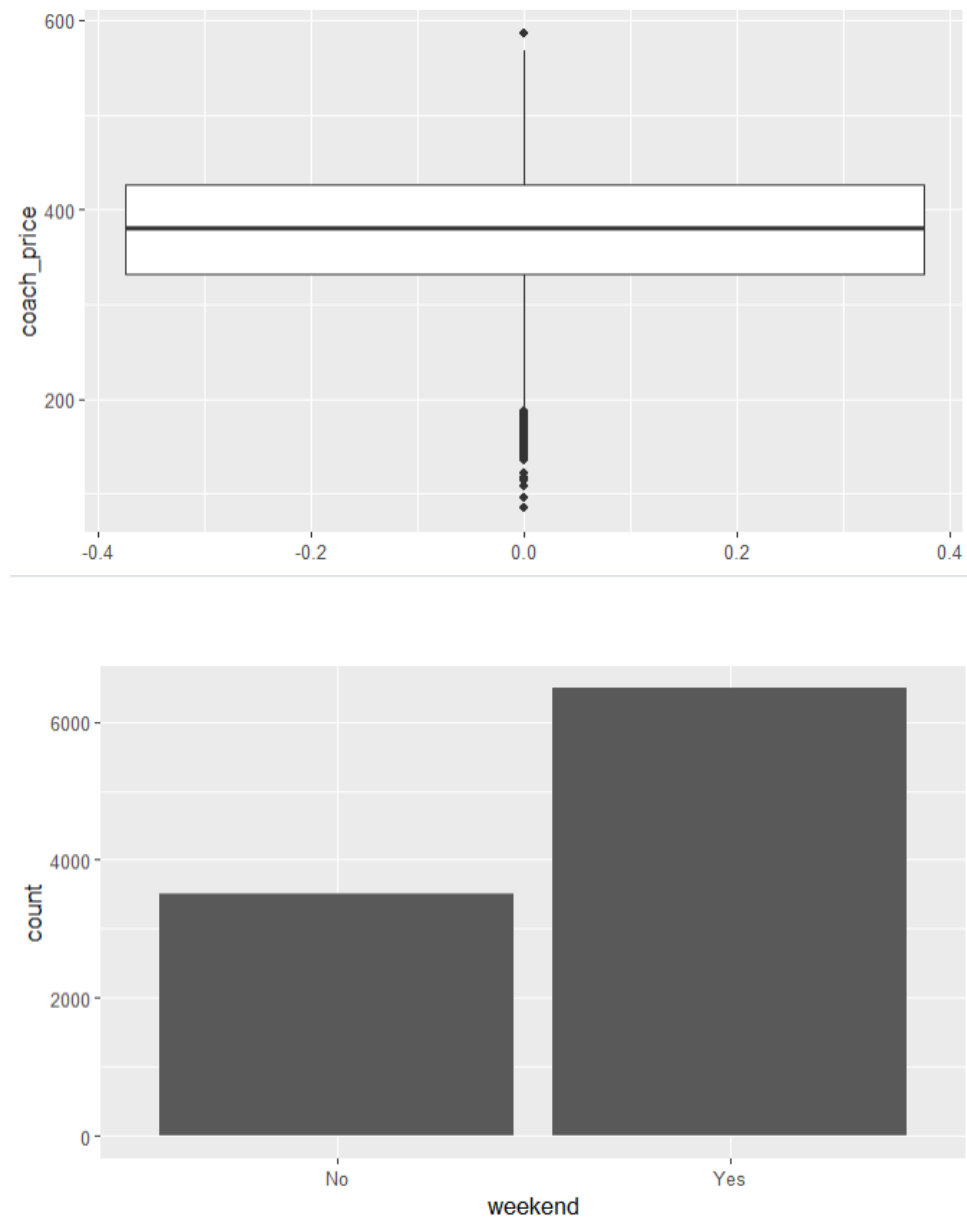
```
1 summary(flight_sample)
```

13.2 Visualization

```
1 # Histogram
2 ggplot(flight_sample,
3       aes(x=coach_price)) +
4   geom_histogram(bins=30)
5
6 # Boxplot
7 ggplot(flight_sample,
8       aes(y=coach_price)) +
9   geom_boxplot()
10
11 # Bar chart
12 ggplot(flight_sample,
13       aes(x=weekend)) +
14   geom_bar()
```

13.3 Output





13.4 Hypothesis Testing

```
1 # Independent t-test
2 t.test(
3   coach_price ~ weekend,
4   data=flight_sample
5 )
```

13.5 Correlation Analysis

```
1 cor(
2   flight_sample[, c(
3     "coach_price",
4     "miles",
5     "hours",
```

```
6     "delay",  
7     "firstclass_price"  
8   )]  
9 )
```

13.6 Linear Regression

```
1 model <- lm(  
2   coach_price ~ miles + hours + delay,  
3   data=flight_sample  
4 )  
5  
6 summary(model)
```

13.7 Logistic Regression

```
1 flight_sample$redeye_binary <-  
2   ifelse(  
3     flight_sample$redeye=="Yes",  
4     1,0  
5   )  
6  
7 log_model <- glm(  
8   redeye_binary ~ hours + coach_price,  
9   data=flight_sample,  
10  family="binomial"  
11 )  
12  
13 summary(log_model)
```

14 Conclusion

This analysis explored several important aspects of airline ticket pricing and flight characteristics using R programming. The study identified relationships between ticket prices, distance, flight duration, delays, and additional flight services. Statistical analysis and visualization techniques helped provide meaningful insights into airline pricing patterns.

15 References

- R Programming Documentation
- ggplot2 Package Documentation
- dplyr Package Documentation